

Ducky Script 3.0 Language Reference

Full Concise Specification & Syntax Guide for Android USB HID Client

1. Overview

This document details the complete language specification for the Ducky Script 3.0 engine embedded within the USB HID Client app. The engine features an asynchronous lexer, AST parser, and coroutine interpreter supporting text payloads, control flow, math expressions, functions, and key state manipulation.

2. Text Typing Commands

Command	Description	Syntax Example
STRING	Sends literal text to the host machine via HID keyboard scan codes.	STRING Hello World!
STRINGLN	Sends literal text followed automatically by an ENTER key press.	STRINGLN echo "Done"

3. Delay Controls

Command	Description	Syntax Example
DELAY	Pauses execution for duration in ms. Accepts expressions or variables.	DELAY 500 DELAY (\$PAUSE * 10)
DEFAULT_DELAY / DEFAULTDELAY	Sets automatic delay (ms) applied after every key press/chord.	DEFAULT_DELAY 100
STRINGDELAY / STRING_DELAY	Sets inter-character delay (ms) between typed characters.	STRINGDELAY 25

4. Key Presses & Combination Chords

Supported Single Keys: ENTER, TAB, ESC, SPACE, BACKSPACE, DELETE, INSERT, PAUSE, PRINTSCREEN, CAPSLOCK, HOME, END, PAGEUP, PAGEDOWN, UP, DOWN, LEFT, RIGHT, F1..F12.

Modifiers: CTRL, ALT, SHIFT, WIN (WINDOWS / GUI / CMD / META / SUPER)

Key Combination Chords:

```
GUI r
CTRL ALT DEL
CTRL SHIFT ESC
ALT F4
```

5. Key Holding & Releasing

Press and hold keys across multiple steps, then release explicitly:

```
HOLD SHIFT
STRING abc
```

```
RELEASE SHIFT
```

6. Variables & Expressions

Variables start with **\$**. Supports arithmetic (+, -, *, /, %) and comparison (==, !=, <, >, <=, >=) operators.

```
VAR $COUNT = 0
VAR $TIMEOUT = 1000
$STEP = ($COUNT + 1) * 2
```

System Constants: `$_CAPSLOCK_ON`, `$_NUMLOCK_ON`, `$_SCROLLLOCK_ON`

7. Control Flow & Loops

If-Else Conditional:

```
IF $STATUS == 1
    STRINGLN Success
ELSE
    STRINGLN Failed
END_IF
```

While Loop:

```
VAR $I = 0
WHILE $I < 5
    STRINGLN Loop iteration
    VAR $I = $I + 1
END_WHILE
```

For Loop:

```
FOR $I = 1 TO 3
    STRINGLN Step $I
END_FOR
```

Repeat Last Action:

```
DOWN
REPEAT 5
```

8. Custom Functions

```
FUNCTION OPEN_RUN_DIALOG
    GUI r
    DELAY 200
    STRINGLN cmd.exe
END_FUNCTION

OPEN_RUN_DIALOG
```

9. Comments & Syntax Diagnostics

- Comments begin with **REM** or **//**.
- The editor parses AST syntax live and underlines errors prior to execution.
- Hardware/mass-storage specific commands (e.g. *ATTACKMODE*, *EXFIL*) raise non-blocking warnings.